Which ML Models Are Actually Used in Industry?

A Literature & Industry Review

Applied Data Science

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1. Most Commonly Used Models in Industry

Despite widespread attention on LLMs and deep learning, industrial ML deployments rely on a much broader set of models. Based on Kaggle's large-scale practitioner surveys and industry reports, three families dominate production environments:

Classical Models (Linear/Logistic Regression, Decision Trees): These remain the most frequently deployed models overall. Their interpretability satisfies regulatory requirements in finance and healthcare , and their low computational cost makes them ideal for high-throughput systems.

Gradient Boosting Ensembles (XGBoost, LightGBM, CatBoost): For structured/tabular data, GBMs are the industrial workhorse . XGBoost and LightGBM dominate in fraud detection, credit scoring, demand forecasting, and click-through rate prediction. In manufacturing, they are the leading choice for predictive maintenance. In retail, LightGBM tops major forecasting competitions and is widely adopted in e-commerce .

Model Family	Key Use Cases	Why Preferred
Linear ,Logistic Regression	Credit scoring, risk models	Interpretability, regulatory compliance
Random Forest , Decision Tree	Fraud detection, churn pred	Balance of accuracy and explainability
XGBoost , LightGBM	Demand forecasting, anomaly detection	Best accuracy on tabular data

Table 1: ML model families and their industrial deployment patterns.

2. Future Outlook: Next 5–10 Years

Classical models will persist but shrink in share. Regulatory demand for explainability ensures linear models stay relevant, especially in finance and insurance . However, as AutoML and SHAP-based explainability tools lower the barrier to adopting complex models, their share of new deployments will gradually decline .

Gradient boosting will remain dominant for tabular data. Deep learning's advantage over GBMs on structured data remains marginal in practice. XGBoost and LightGBM will continue to be the default for most business ML tasks for at least another decade .

LLMs will transform language-heavy domains. The shift from general-purpose LLMs to domain-specific fine-tuned models is already underway in 2026 . Legal tech, healthcare documentation, and software development will see the strongest disruption . Autonomous LLM-based agents will increasingly replace rule-based pipelines in customer-facing and back-office systems.

In short, the next decade will produce a clear bifurcation: *gradient boosting dominates structured data* , while *LLMs and deep learning dominate language and vision.* The winners will be organizations that match the right model family to each problem rather than chasing a single paradigm .

References

[1] Kaggle. *State of Data Science & Machine Learning Survey (over the years.* https://www.kaggle.com/code/paultimothymooney/2021-kaggle-data-science-machine-learning-survey

[2] Vidora. *Kaggle Survey: Biggest Challenges Deploying ML.* https://www.vidora.com/general/kaggle-survey

[3] JetBrains. *The State of Data Science 2024*. Dec 2024. https://blog.jetbrains.com/pycharm/2024/12/the-state-of-data-science

[4] Analytics Insight. *Exploring Data Science Tools: What's Hot in 2024*. Oct 2024. https://www.analyticsinsight.net

[5] Chawla, A. *Top Gradient Boosting Methods*. Daily Dose of DS, Oct 2025. https://blog.dailydoseofds.com/p/top-gradient-boosting-methods

[6] Raschka, S. *State of LLMs 2025*. Jan 2026.
https://magazine.sebastianraschka.com/p/state-of-llms-2025

[7] Turing.com. *Top LLM Trends 2025*. 2026.
https://www.turing.com/resources/top-llm-trends

[8] Kalaskar et al. *AI for Predictive Maintenance in Steel Industry*. arXiv:2405.12785, 2024.

[9] Anonymous. *Retraining Frequency in Retail Demand Forecasting*. arXiv:2505.00356, 2025.

[10] Anonymous. *ML Analysis: Medical Conditions and COVID-19*. arXiv:2112.12901, 2021.

[11] Anonymous. *CatBoost for Loan Risk Assessment*. arXiv:2106.07954, 2021.